

AL: Agentic Lender

Securing a mortgage or home equity line of credit (HELOC) is traditionally slow, paper-heavy, and opaque. Financial institutions face significant underwriting overheads, while downstream loan activities and secondary-market sales require costly manual reconciliations between banks, title registries, and brokers.

FinTech pioneers such as Figure (<https://www.figure.com/>) have demonstrated that placing loan origination on a distributed ledger cuts settlement times from weeks to minutes while creating a single, tamper-proof source of truth. By combining automated property valuation models (AVM) with multi-agent AI committees and smart contracts, lending institutions can automate applicant creditworthiness, appraise fair collateral values, and mint verifiable digital credit agreements.

Your team of four will develop a bespoke prototype system called **AL (Agentic Lender)**. The platform connects retail borrowers, accredited mortgage brokers, and bank underwriting officials within a modern lending ecosystem.

System Overview

The platform has four distinct subsystems, aligning with a 4-person team structure:

- **AL-Mobile (Mobile Client):** Native Android app for retail borrowers to calculate borrowing capacity, submit property specs, check fair property valuations, and track their on-chain application status.
- **AL-Web (Multi-View Portal):** React web application for mortgage brokers and bank underwriting officials to manage loan pipelines, inspect agent reasoning trails, and execute on-chain loan approvals.
- **AL-Core & Ledger (Backend, Database & Blockchain):** Centralized Firebase database for state/auth, alongside an EVM-compatible smart-contract layer that mints tokenized loans and logs immutable audit hashes.
- **AL-AI Engine (Backend Engine):** Python service (e.g. using Flask) combining supervised machine learning for property valuation with an agentic AI committee that deliberates on credit suitability.

Features

1. AL-Mobile (Mobile Client – Android / Kotlin)

- **Register/Login:** Secure client sign-in via email/password or social authentication using Firebase Auth.
- **Affordability & Loan Application:** Dynamic forms capturing household annual gross income, existing debt liabilities, deposit, and requested mortgage term.
- **Fair-Price Property Checker:** Borrowers enter target property specifications (location/county, square meters/footage, bedrooms, property type, and BER/energy rating) to query an automated valuation estimate from the ML model.
- **AL-AI Interface & Status:** Visual presentation of estimated Loan-to-Value (LTV), Debt-to-Income (DTI), approval likelihood, and the current lifecycle state of their digital loan note.
- **Broker / Official Contact:** In-app messaging to communicate directly with the assigned mortgage broker or bank lending officer.
- **Payments:** Fast-track appraisal processing or application verification fee integrated via Stripe Sandbox or PayPal Sandbox.
- **Support & Reviews:** Automated customer support forms and in-app rating/review submission.

2. AL-Web (Multi-View Portal – React)

Developed as a React application containing distinct authenticated views:

- **Role-Based Access:** Independent Mortgage Brokers (restricted to viewing and managing only their referred client files) and Bank Underwriting Officers (portfolio-wide oversight with sign-off authority).
- **Origination Pipeline View:** Centralised queue displaying incoming loan applications, categorised by LTV thresholds, risk tiers, and lifecycle status (Applied, AI Deliberated, Underwriter Approved, Funded, Closed).
- **Loan File & Collateral Assessment View:** Detailed dossier of individual applications comparing the seller's asking price against the ML valuation prediction, with complete borrower DTI breakdowns.
- **AL-AI Audit & Deliberation Console:** Transparent interface displaying the multi-agent committee reasoning logs, agent consensus, and individual decision trails.
- **Macro Stress-Testing Simulator:** Interactive graphical interface enabling bank managers to simulate central bank interest rate shocks (+100 to +300 bps) or market downturns against active applicant portfolios.
- **On-Chain Settlement & Decision Export:** Allows authorised bank officials to sign off on an application, triggering smart-contract loan minting and generating downloadable PDF/CSV audit packs.

3. AL-Core, Database & Blockchain (Firebase & EVM Smart Contracts)

- **Centralised Data Store:** Cloud Firestore maintaining user profiles, document metadata, broker assignments, and real-time application states.
- **Loan Tokenization Contract (Solidity):** Represents approved mortgages/HELOCs as digital loan notes (ERC-721 or stateful smart contract) containing principal, APR, term, and borrower public identifier.
- **Immutable Audit & Decision Registry:** Directly commits a cryptographic SHA-256 hash of the final credit memo, valuation data, and underwriter approval onto the blockchain to ensure verifiable, tamper-proof lending records.
- **Execution Environment:** Deployed locally using Hardhat/Anvil (via Docker Compose) or to a public EVM testnet (Polygon Amoy or Arbitrum Sepolia).

4. AL-AI Engine (Predictive ML & Multi-Agent Committee)

A Python service linking AL-Mobile and AL-Web, combining supervised predictive modelling and multi-agent workflows:

- **Automated Property Valuation Model (ML):** A trained regression model using modern tabular housing datasets (e.g., *Ireland House Properties 2024 / Daft.ie*, *Irish Property Price Register*, or *UK House Price Prediction Dataset*) to establish fair property valuations.
- **Agentic AI Architecture (CrewAI, AutoGen, or LangGraph):**
 - **Affordability Agent:** Evaluates net disposable surplus, stress-tested repayments, and regulatory borrowing caps.
 - **Valuation Agent:** Compares asking prices against the supervised valuation model, flags overvalued collateral, and computes verified Loan-to-Value (LTV).

- **Market & Macro Researcher Agent:** Uses external search tools (e.g., YahooFinanceNewsTool, Serper, or Web Search) to pull current central bank base rates and prevailing mortgage benchmark trends.
- **Underwriting Strategist Agent:** Deliberates on upstream outputs, determines final lending decisions (Approve, Conditionally Approve, or Decline), and generates an audit-ready credit memorandum.
- **Model Diversity & Local Inference:** Integration combining at least two cloud-based LLM APIs (e.g., Groq, ChatGPT, Gemini, or Claude) with a locally hosted LLM executed via **Ollama** (e.g., Llama 3, Mistral).

Technology Stack Summary

Subsystem	Core Technology	Primary Responsibility
AL-Mobile	Android Studio, Kotlin	Borrower app, property input UI, loan status, payment flows
AL-Web	React (React Router, Tailwind CSS)	Broker & underwriter dashboard, audit trails, on-chain execution
AL-Database & API	Firebase (Firestore/Auth), Python (Flask/FastAPI)	User management, application routing, ML/agent dispatching
Blockchain / Ledger	Solidity, Hardhat, Ethers.js / Web3.py	Digital loan registry, audit trail hashing, state lifecycle
Agentic Framework	CrewAI, AutoGen, or LangGraph	Multi-agent committee deliberation and external market search tools
Predictive ML	Scikit-learn or TensorFlow / Keras	Supervised regression model for fair property valuation (AVM)
LLM Inference	Ollama (Local) + Cloud APIs (Groq / Gemini)	Dual cloud + private local multi-model consensus
Payments & Tooling	Stripe/PayPal Sandbox, GitHub, Jira	Appraisal fee payments, Git code management, project tracking

Custom Feature Requirement

Each group of four must implement at least one unique custom feature in AL-Mobile, AL-Web, AL-AI Engine, or the Blockchain layer. Potential options:

- **Document OCR Pipeline:** Android camera or Web document upload module using Tesseract/OCR to parse payslips and extract applicant income and deductions automatically.
- **Green Mortgage Smart Incentive:** Smart contract logic that inspects a property's Building Energy Rating (BER/EPC) and automatically deducts 25 bps from the contract's baseline APR for energy-efficient homes.
- **First-Time Buyer Explainer Bot:** Conversational in-app assistant fine-tuned on regional mortgage terminology, government first-home schemes, and loan closing costs.
- **Neighbourhood Amenity Scoring Agent:** Agentic integration with mapping APIs (Google Maps / OpenStreetMap) to score property proximity to schools, transit, and amenities as a risk modifier.

References & Starting Points

Datasets

- Ireland House Properties Dataset 2024 (Daft.ie listings with BER ratings and floor area): <https://www.kaggle.com/datasets/adnankhalid007/ireland-house-properties-dataset-2024>
- Irish Property Price Register (PPR): <https://www.kaggle.com/datasets/fionnhughes/property-price-register>
- UK House Price Prediction Dataset: <https://www.kaggle.com/datasets/swarupsudulaganti/uk-house-price-prediction-dataset-2015-to-2024>
- Kaggle Loan Prediction Dataset: <https://www.kaggle.com/datasets/altruistdelhite04/loan-prediction-problem-dataset>

Blockchain & Web3

- OpenZeppelin Smart Contract Library (ERC-721 & AccessControl): <https://docs.openzeppelin.com/contracts>
- Hardhat EVM Development Environment: <https://hardhat.org/>
- Web3.py (Python Blockchain Interface): <https://web3py.readthedocs.io/>

Agent Frameworks & LLM Hosting

- CrewAI Documentation: <https://docs.crewai.com/>
- AutoGen Multi-Agent Framework: <https://microsoft.github.io/autogen/>
- Ollama (Local LLM Execution): <https://ollama.com/>
- Groq Developer Console: <https://console.groq.com/>
- Machine Learning Basics (Regression): <https://machinelearningmastery.com>